

4.2 Encoder Truth Table:

D_0	D_1	D_2	D_3	x	y	<u>Question:</u>
1	0	0	0	0	0	How do you say
0	1	0	0	0	1	$y = D_1 + D_3$, when
0	0	1	0	1	0	it is actually
0	0	0	1	1	1	$D_0' D_1 D_2' D_3' +$
						$D_0' D_1' D_2' D_3$?

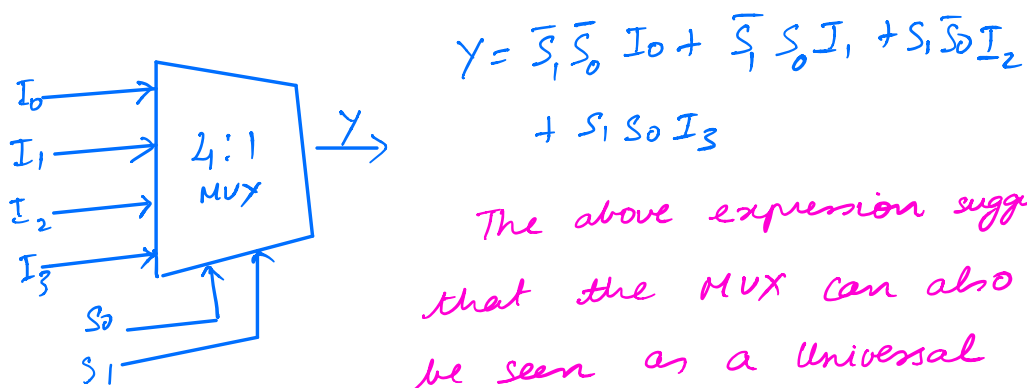
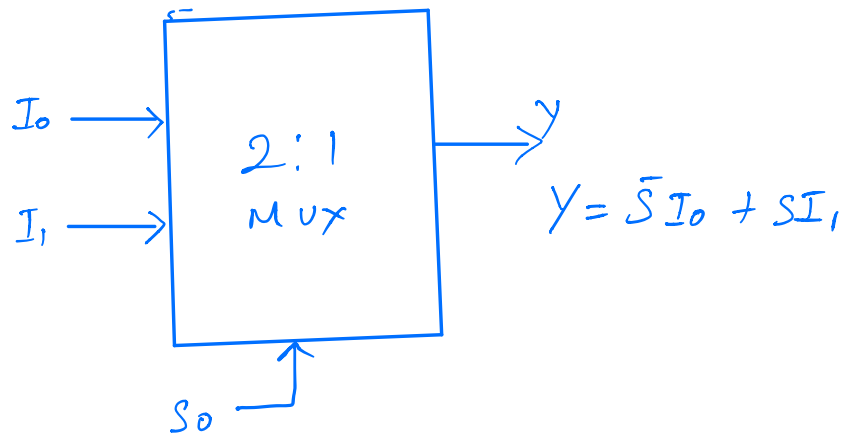
Answer: The fact that $D_1 = 1 \Rightarrow D_0 = D_2 = D_3 = 0$
 and $D_3 = 1 \Rightarrow D_0 = D_1 = D_2 = 0$ would make
 $y = D_1 + D_3$ as simply $D_1 = 1$ implies automatically
 other entries are zero in the input columns.

MULTIPLEXER: A logical circuit that performs
 selection or sampling function.

Given m , select lines and 2^m input lines
 selects one amongst the 2^m input lines
 based on the select input values.

Transfers the value of the selected input

line to the output



The above expression suggests that the MUX can also be seen as a Universal

Boolean Function Realizer or Implementer

For any Boolean function $f(x, y, z) = f_0(0,0,0) + f_1(0,0,1) + \dots + f_7(1,1,1)$

Select inputs

And hence $f(x, y, z) = f_1(x, y, 0) + f_2(x, y, 1)$
if z alone is the select line

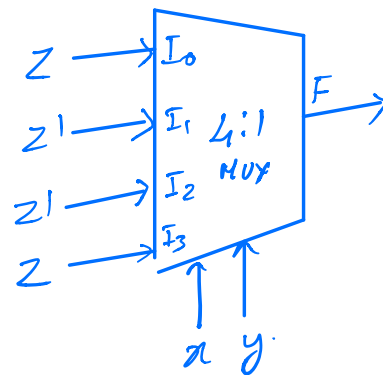
$$\text{or } f(x, y, z) = f_1(x, 0, 0) + f_2(x, 0, 1) + f_3(x, 1, 0) + f_4(x, 1, 1).$$

For example: $F(x, y, z) = \sum(1, 2, 4, 6, 7)$

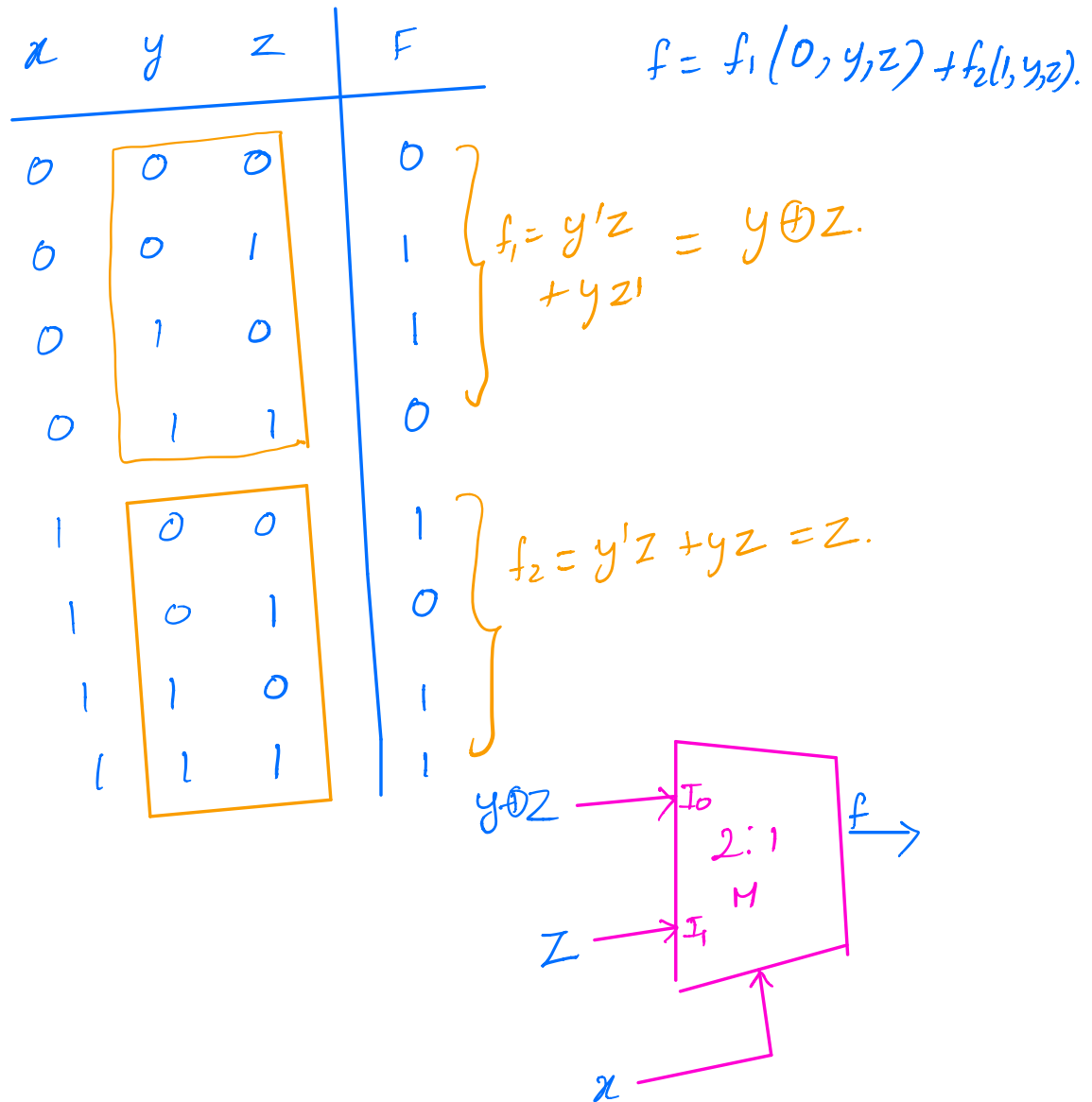
	x	y	z	F
m ₀	0	0	0	0
m ₁	0	0	1	1
m ₂	0	1	0	1
m ₃	0	1	1	0
m ₄	1	0	0	1
m ₅	1	0	1	0
m ₆	1	1	0	1
m ₇	1	1	1	1

$$F = f_1(0, 0, z) + f_2(0, 1, z) + f_3(1, 0, z) + f_4(1, 1, z).$$

$\therefore$ Realized through 4:1 MUX with x and y as select line



Q: Can you realize the above with a 2:1 MUX



Question: What is the difference between Universal Boolean function realization via Decoder and via MUX?